

Roshni Poddar

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Experience

July 2024
Present

Microsoft

Software Engineer

Windows Empower Team Improving the core user experience of the Windows built-in screen reader, Narrator, with Microsoft Office apps [C++, C#]. Also conducting qualitative user research to understand how beginners learn screen readers and identify key friction points in Narrator's usage.

Azure Local Team Co-developed the new Arc Sites feature on Azure Portal that enables over 1k clients to model on-premises environments and access centralized edge infrastructure monitoring [React.js, TypeScript, Playwright].

Jan 2022
July 2024

Microsoft Research

Research Fellow / Advisors: *Dr. Manohar Swaminathan, Dr. Mohit Jain, Dr. Pratyush Kumar*

1. Engineered an Android app (SEEDS) to facilitate accessible content delivery between visually-impaired teachers and students. SEEDS is deployed at schools in 3 Indian states and expanding nationwide.

> Implemented an Interactive Voice Response system using FastAPI, enabling children to independently interact with accessible stories, quizzes, and audio games.

> Created React.js website for educators to manage educational content and track student engagement.

> Led a two-month study with 29 visually-impaired children and 5 teachers to evaluate the effectiveness and identify potential areas for refinement. Published [paper](#) at COMPASS'24.

2. Engineered a sign language-based Android quiz app (SignIt) co-designed with the Deaf community to facilitate learning sign language and English through play.

> SignIt offers single and multiplayer play modes along with guidance to ensure quiz creators' upper body visibility before recording.

> Presented findings from a user study with 20 Deaf participants at [poster](#) at ASSETS'23; full [study](#) published in the ITU'23 journal.

3. Enhanced user experience for an accessible video-conferencing web app for mixed hearing groups in collaboration with researchers at the University of Washington.

> Conducted user studies with 18 Deaf and Hard of Hearing participants, 6 interpreters, and 10 hearing individuals. Published [paper](#) at ASSETS'23.

August 2020
Dec 2021

AI4Bharat

Research Intern / Advisor: *Dr. Pratyush Kumar*

[GitHub](#)

1. Trained and evaluated machine learning models to convert sign language in videos into text captions.

> Achieved 88.44% accuracy on 4287 videos and 263 word signs using transformer model architecture.

> Conducted 15 semi-structured interviews and 131 survey responses to understand the challenges faced by the employed Deaf community in India. Published [paper](#) at INTERACT'23.

Education

Aug 2018
May 2022

PES UNIVERSITY

B.Tech, Computer Science & Engineering

CGPA: 9.17 out of 10

Bangalore, India

Honours and Awards

Microsoft Hackathon, 2022. Won third place in Hack 2 enable and Hack for Society categories among 10,019 entries

Microsoft Hackathon, 2024. Led a 5-member team to win third place in the Hack4Inclusion challenge among 20,249 entries

Invited speaker at Carnegie Mellon University, 2024. Presented on *Empowering diverse learning for children with vision impairments*, showcasing innovative strategies for inclusive education.

Prof. CNR Rao Scholarship. Received merit scholarships for academic excellence during undergraduate studies.

Skills

Programming Languages

C++, C#, JavaScript, TypeScript, HTML5, CSS, Kotlin (Android development), Python

Frameworks & Technologies

React, Next.js, Playwright, FastAPI, Flask, Express.js

Cloud & Database Technologies

Azure Cloud Services, MongoDB, MySQL

Accessibility

Web accessibility, Android accessibility, Apple accessibility APIs, screen readers